

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION**Annual Examination 2026****MATHEMATICS — SSC-I (9th Grade)****Total Marks: 65****Time Allowed: 2 Hours 30 Minutes****Version: 1****General Instructions:**

1. **Section A** (MCQs): Attempt **ALL 12** questions. Each carries **1 mark**. Time: 15 minutes.
2. **Section B** (Short Questions): Attempt **11**, choosing main **OR** alternative. Each **3 marks**.
3. **Section C** (Long Questions): Attempt **4**, choosing main **OR** alternative. Each **5 marks**.
4. Use black or blue ink only. Pencil is not allowed except for diagrams and number lines.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)*Encircle the correct option. All questions are compulsory.*

#	Question	A	B	C	D	Answer
1	Which of the following is an irrational number?	$5\sqrt{8}$	$\sqrt{16}$	$\sqrt{5}$	0.25	ⒶⒷⒸⒹ
2	The property $a(b + c) = ab + ac$ is called:	Commutative	Associative	Distributive	Identity	ⒶⒷⒸⒹ
3	The set of real numbers R is the union of:	Naturals & Wholes	Integers & Primes	Rationals & Irrationals	Evans & Odds	ⒶⒷⒸⒹ
4	The additive inverse of -7 is:	0	$-1/7$	$1/7$	7	ⒶⒷⒸⒹ
5	In the expression $\sqrt[3]{25}$, the number 25 is called:	Radical	Index	Radicand	Exponent	ⒶⒷⒸⒹ
6	The value of a^0 (where $a \neq 0$) is:	0	1	a	Undefined	ⒶⒷⒸⒹ
7	Scientific notation of 0.000052 is:	5.2×10^{-5}	52×10^{-6}	5.2×10^{-4}	0.52×10^{-4}	ⒶⒷⒸⒹ
8	If $\log_2 64 = y$, then y equals:	4	5	6	7	ⒶⒷⒸⒹ
9	The characteristic of $\log 456.7$ is:	0	1	2	3	ⒶⒷⒸⒹ
10	$\log_b (m \times n)$ equals:	$\log m + \log n$	$\log m - \log n$	$\log m \cdot \log n$	$n \log m$	ⒶⒷⒸⒹ
11	The base of the natural logarithm is:	10	2	π	e	ⒶⒷⒸⒹ
12	The value of $\log_b 1$ is:	0	1	b	Undefined	ⒶⒷⒸⒹ

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt **11** questions. Choose either the main question **OR** its alternative.

Q.	Question	Marks	OR	OR Question	Marks
2(i)	State any five basic properties of real numbers with respect to addition. Give the general form of each.	3	OR	Define rational and irrational numbers. Write two examples of each.	3
2(ii)	Identify the property used: (a) $5 \times 7 = 7 \times 5$ (b) $3 + (4 + 2) = (3 + 4) + 2$ (c) $8 \times 1 = 8$	3	OR	State the Trichotomy property for real numbers and verify it for $a = -2$ and $b = 5$.	3
2(iii)	Simplify: (a) $81^{1/4}$ (b) $8^{2/3}$ (c) $16^{-1/2}$	3	OR	Write as a single radical and simplify: (a) $\sqrt[3]{6 \cdot \sqrt[3]{5}}$ (b) $\sqrt{20} \div \sqrt{5}$	3
2(iv)	Use laws of exponents to simplify: $(a^{-2} \cdot a^5) \div a^2$	3	OR	Write $x^{3/4}$ as a radical expression and simplify if possible.	3
2(v)	Find the value of x if $\sqrt{3x + 1} = 5$.	3	OR	Simplify: $(2^3 \cdot 2^4) \div 2^5$	3
2(vi)	Convert to scientific notation: (a) 7,340,000 (b) 0.000089 (c) 125.6	3	OR	Convert to standard notation: (a) 6.7×10^4 (b) 5.43×10^{-3} (c) 2.0×10^6	3
2(vii)	Evaluate without log tables: (a) $\log_5 125$ (b) $\log_2(1/8)$ (c) $\log_{10} 10000$	3	OR	Find x : (a) $\log_3 x = 4$ (b) $\log_x 16 = 2$ (c) $\log_4 64 = x$	3
2(viii)	State the four laws of logarithms with their general form.	3	OR	Differentiate between common logarithm and natural logarithm (any three differences).	3
2(ix)	If $\log 2 = 0.3010$ and $\log 3 = 0.4771$, find: (a) $\log 6$ (b) $\log 12$ (c) $\log 18$	3	OR	Find the characteristic of: (a) $\log 67.3$ (b) $\log 0.00892$ (c) $\log 5$	3
2(x)	Use laws of logarithms to expand: $\log(x^3 y^2 \div z)$	3	OR	Combine into a single logarithm: $2 \log x - 3 \log y + \log z$	3
2(xi)	If $\log 2 = 0.3010$, find the number of digits in 2^{20} .	3	OR	An earthquake measured 6.5 on the Richter scale. Another measured 7.5. How many times stronger is the second?	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Attempt **4** questions. Choose either the main question **OR** its alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	Use properties of exponents to simplify. Show all steps. Assume variables are positive. $(a^{-1/2} \cdot b^{1/3})^6 \cdot (a^2 \cdot b^{-1})$	5	OR	Use laws of exponents to show that: $(x^p/x^q)^{p+q} \cdot (x^q/x^r)^{q+r} \cdot (x^r/x^p)^{r+p} = 1$	5
3(ii)	(a) Represent each on the number line: (i) -3 (ii) $7/4$ (iii) $\sqrt{5}$ (iv) $-2 < x < 3$. (b) Identify the property: (i) $(x + y) + z = x + (y + z)$ (ii) $5 \cdot (1/5) = 1$	5	OR	Define: (a) closure (b) associative (c) distributive property of real numbers. Verify each using $a = 3$, $b = 4$, $c = 5$.	5
3(iii)	Prove the product law of logarithms: $\log_b(mn) = \log_b m + \log_b n$. Using $\log 2 = 0.3010$ and $\log 3 = 0.4771$, evaluate $\log 72$.	5	OR	Prove the quotient law of logarithms: $\log_b(m/n) = \log_b m - \log_b n$. Using $\log 2 = 0.3010$ and $\log 7 = 0.8451$, evaluate $\log(14/8)$.	5
3(iv)	The Kansu (China) earthquake of 1920 measured 8.5 on the Richter scale. The Tokyo (Japan) earthquake of 1923 measured 7.8 on the same scale. How many times stronger was the 1920 earthquake than the 1923 earthquake?	5	OR	Find the number of digits in 5^{37} , given that $\log 5 = 0.6990$.	5

Invigilator's
Signature

Candidate's
Signature

Chapters Covered: Unit 1 — Real Numbers | Unit 2 — Logarithms

— End of Question Paper —